

Week 9

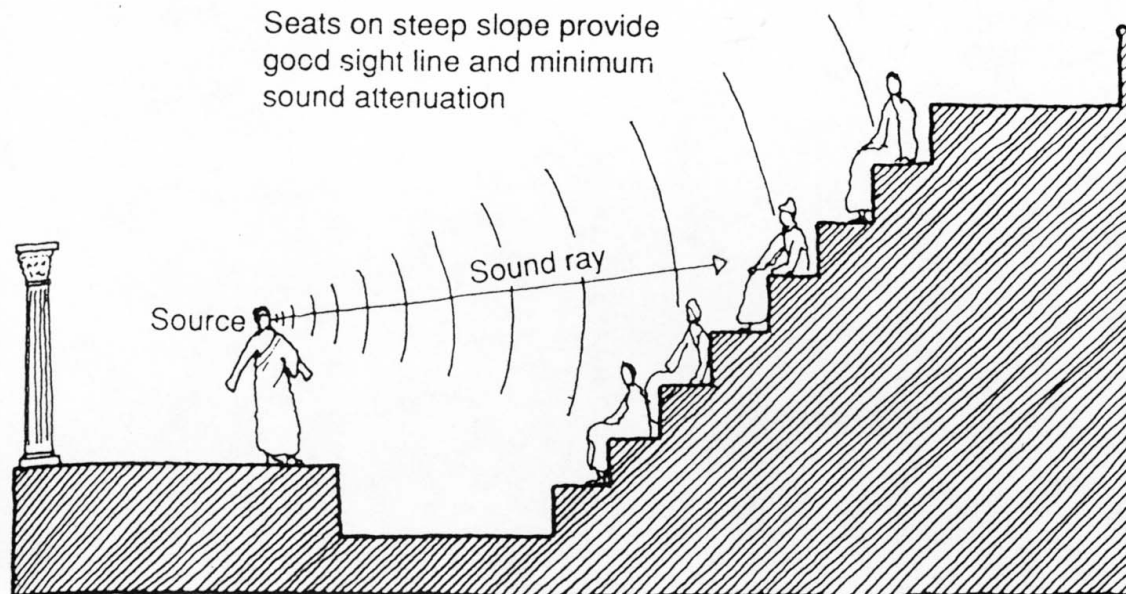
Introduction to Architectural Acoustics

Spring 2021
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College of Architecture and Design
Lawrence Technological University

Historic Overview

Greek Theatre

- Open air
- Direct sound path
- No sound reinforcement
- Minimal reverberation

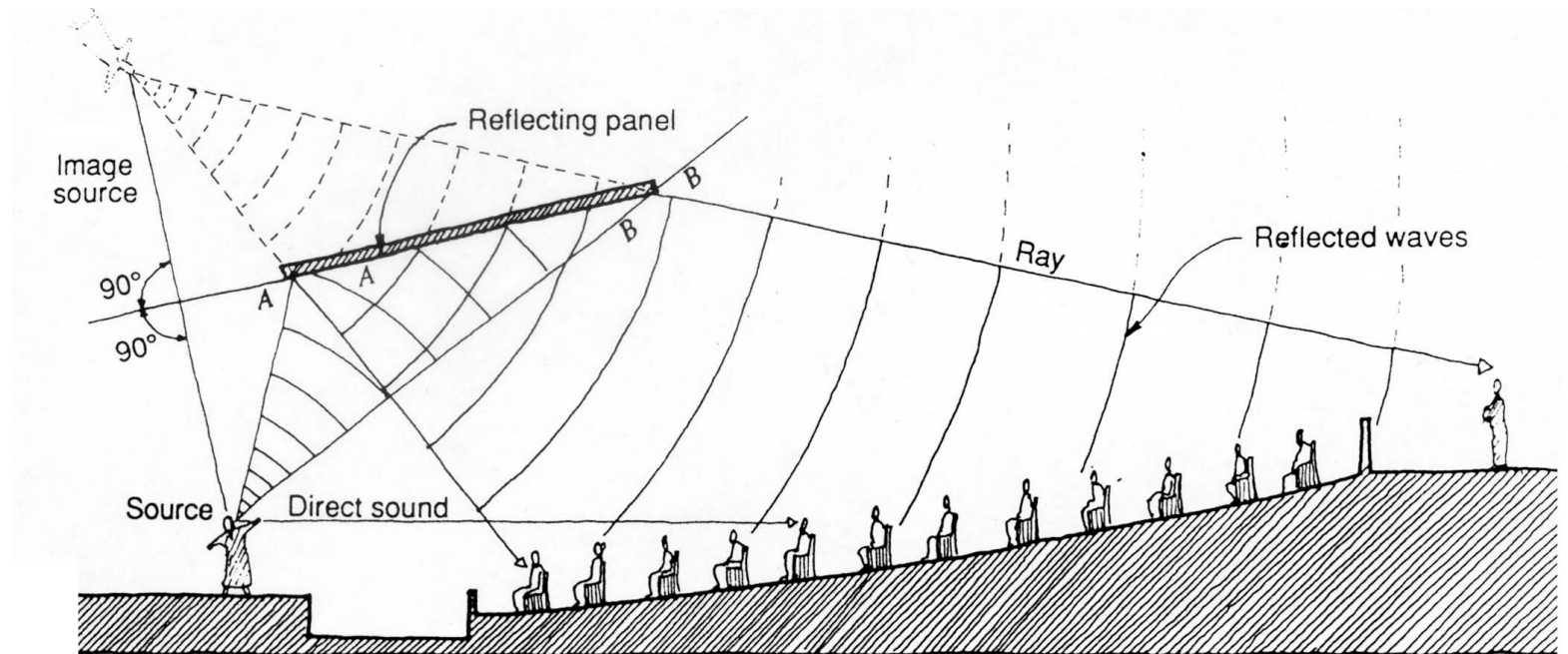


Historic Overview

1st Century AD

Vitruvius: “10 Books of Architecture”

- Sound reinforcement
- Reverberation



Historic Overview

Today

Research to improve conditions for

- Industrial noise
- Hearing risks
- Construction noise
- Public health

Architect's Role

Source

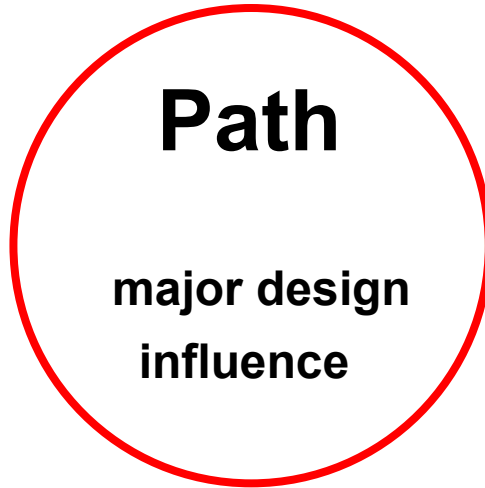
slight

Path

major design
influence

Receiver

primarily interest



Acoustical Design

**“Proper acoustical planning
eliminates many acoustical problems before
they are built”**

Lee Irvine

Acoustical Design Relationship

Site

Location

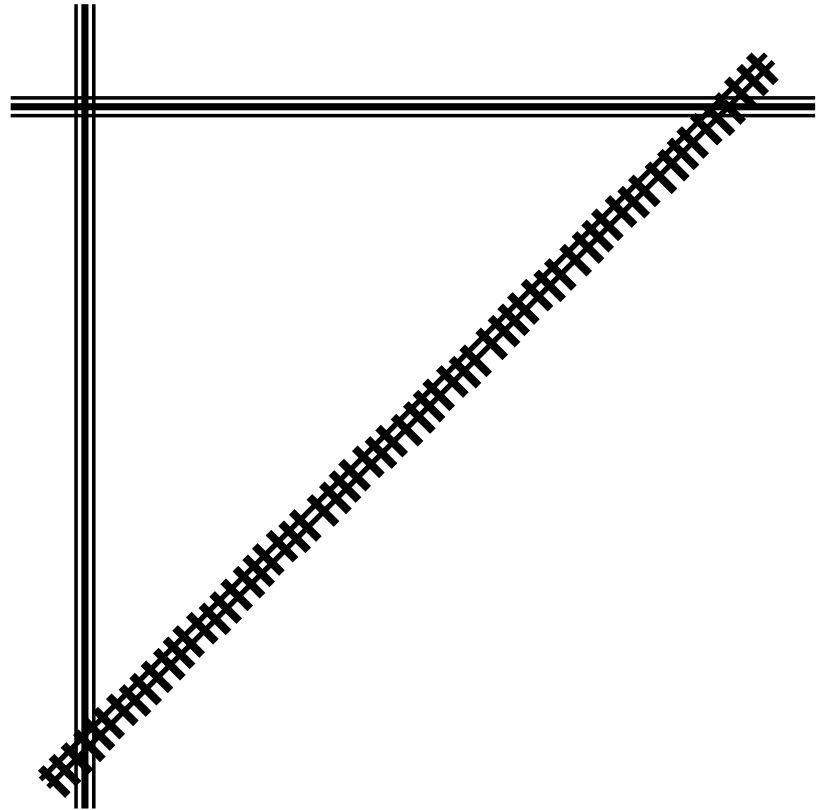
Orientation

Planning

Internal Layout

Site

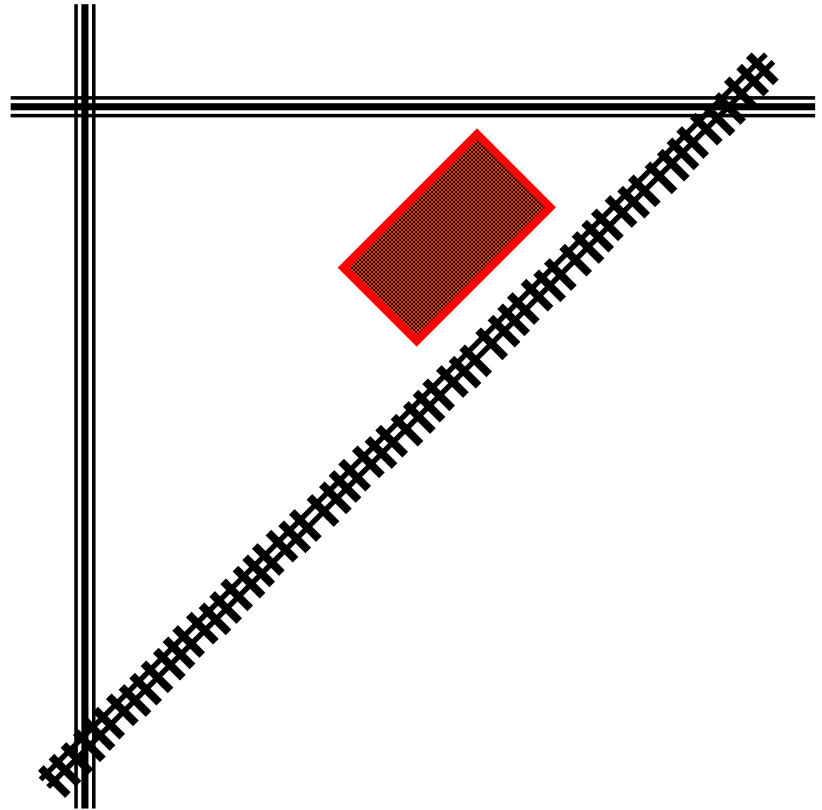
- Match site to application
- Match application to site



Site

Factory:

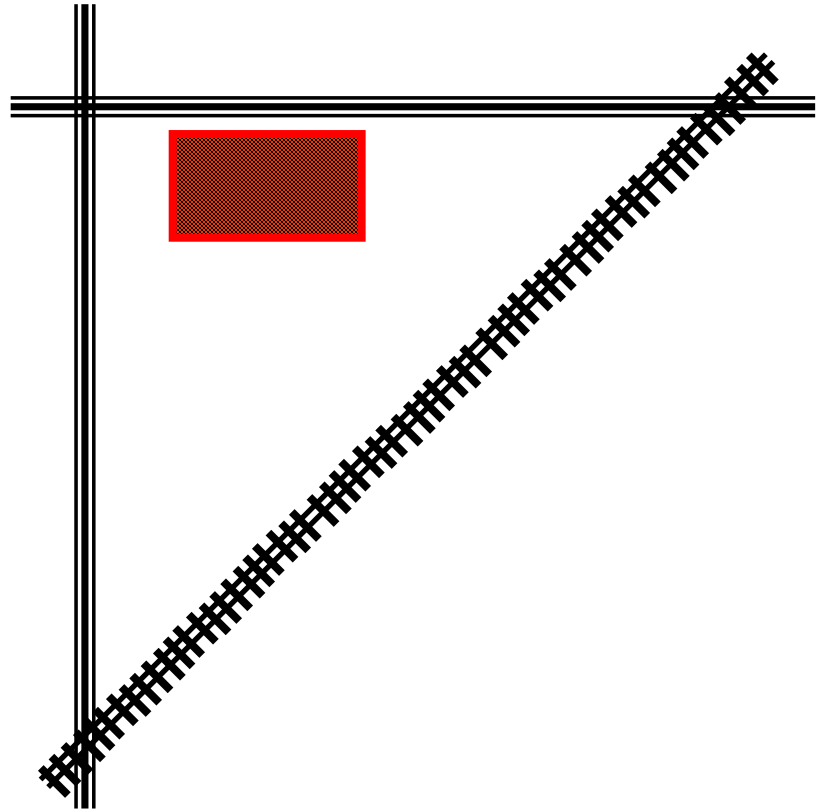
- Close to RR/Hwy
- Seismic



Site

Rest Home:

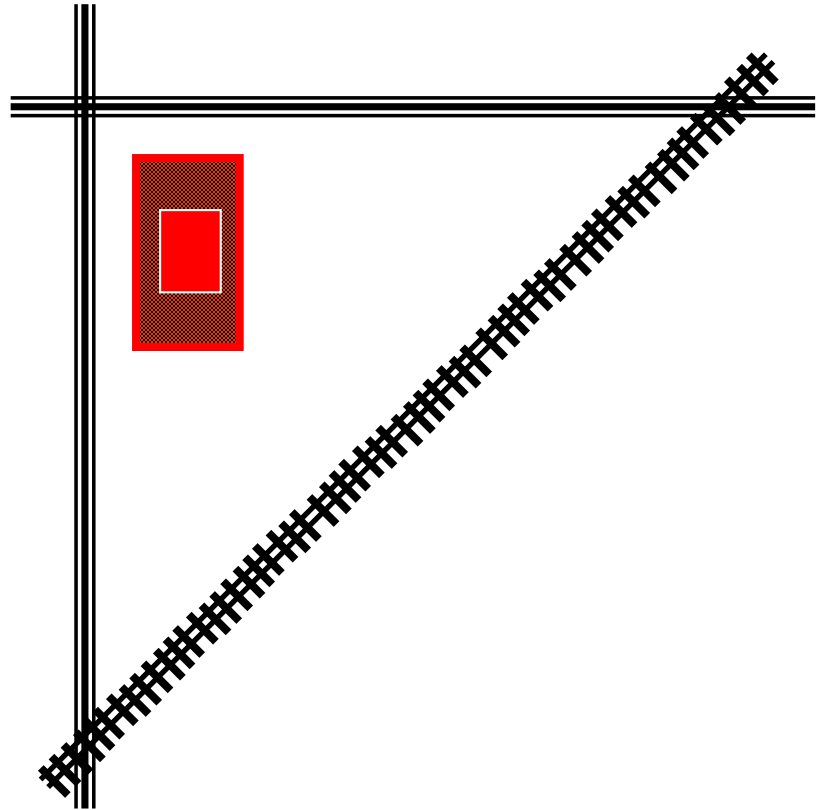
- Traffic Noise
- Outdoor Use
- Contact/Isolation



Site

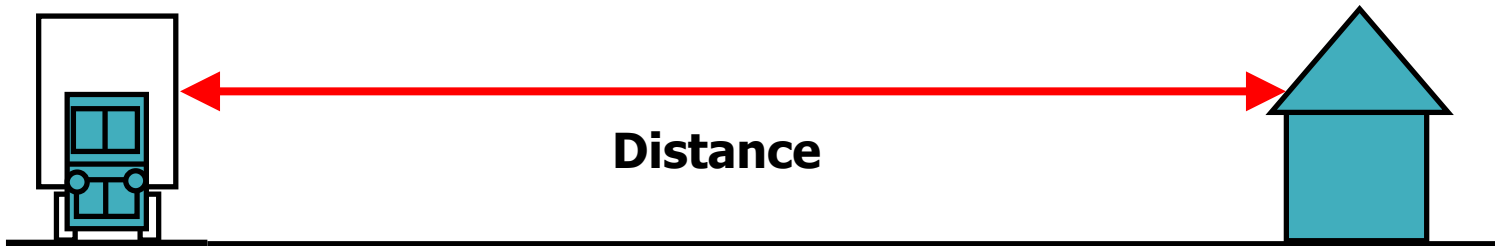
Concert Hall:

- Use building as isolator
- Distance from noise



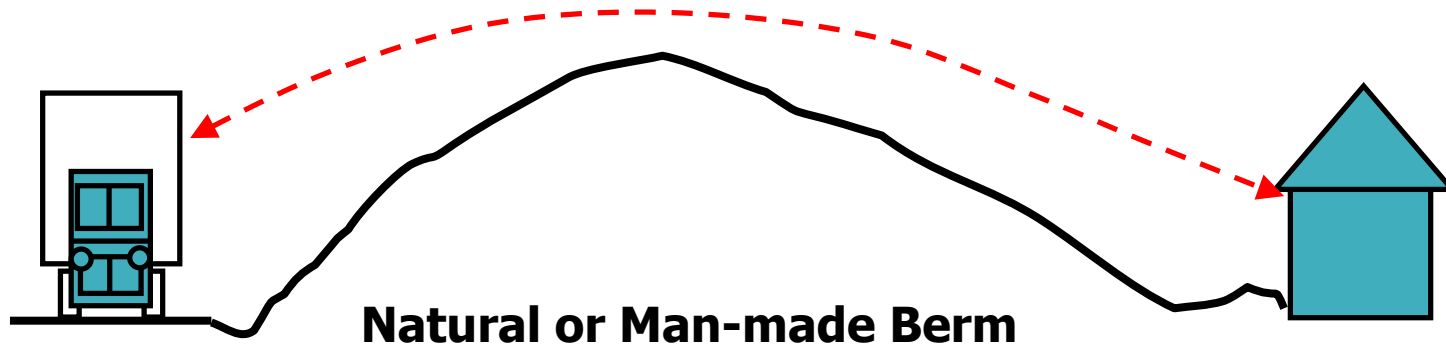
Location

Take advantage of distance/barriers



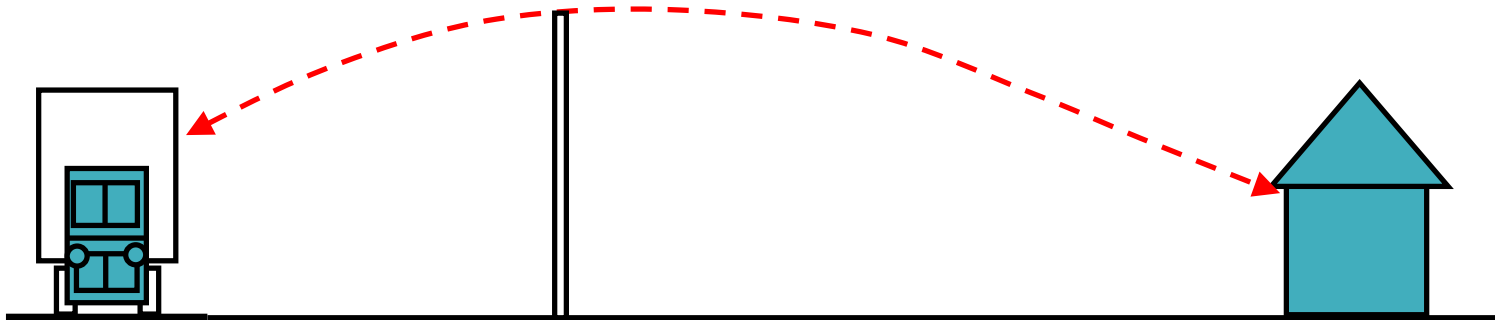
Location

Take advantage of distance/barriers



Location

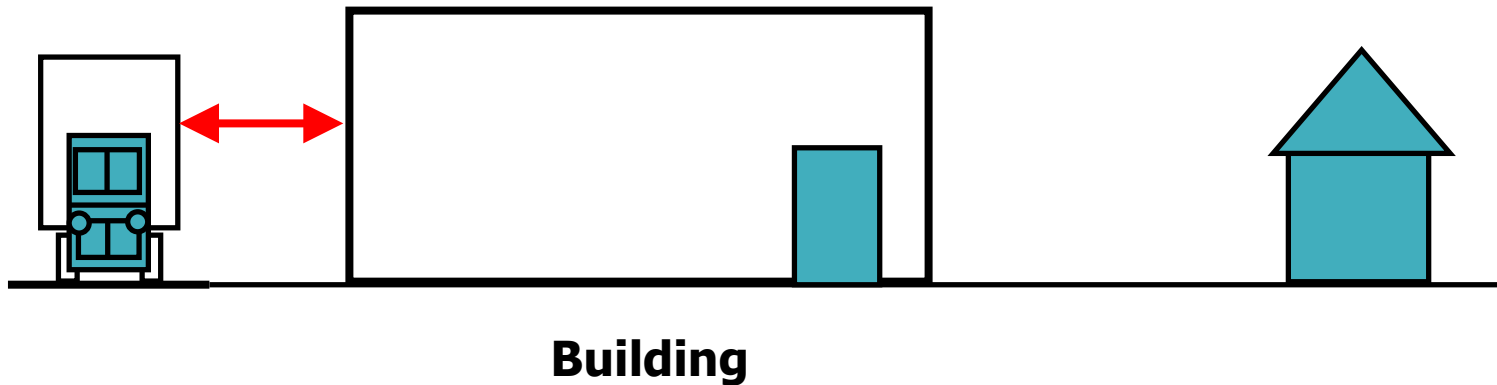
Take advantage of distance/barriers



Acoustical Barriers

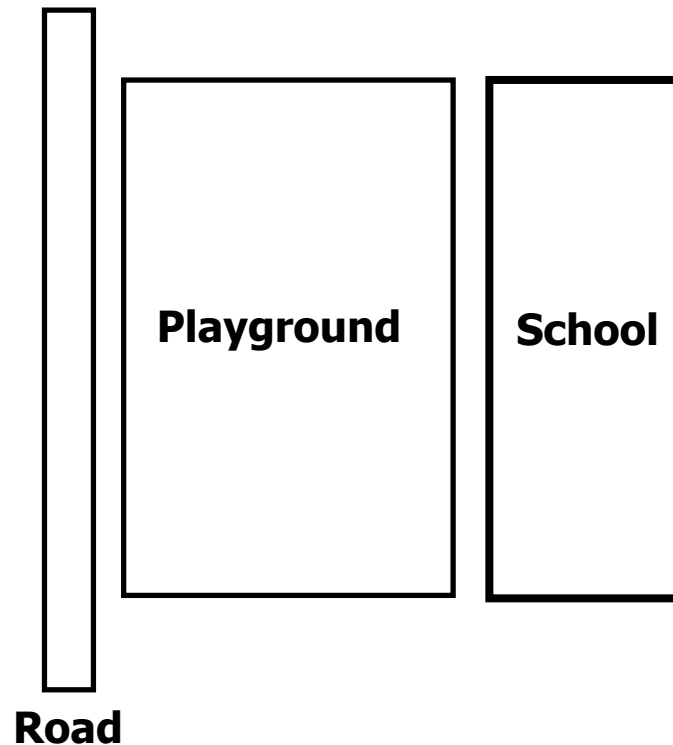
Location

Take advantage of distance/barriers



Orientation

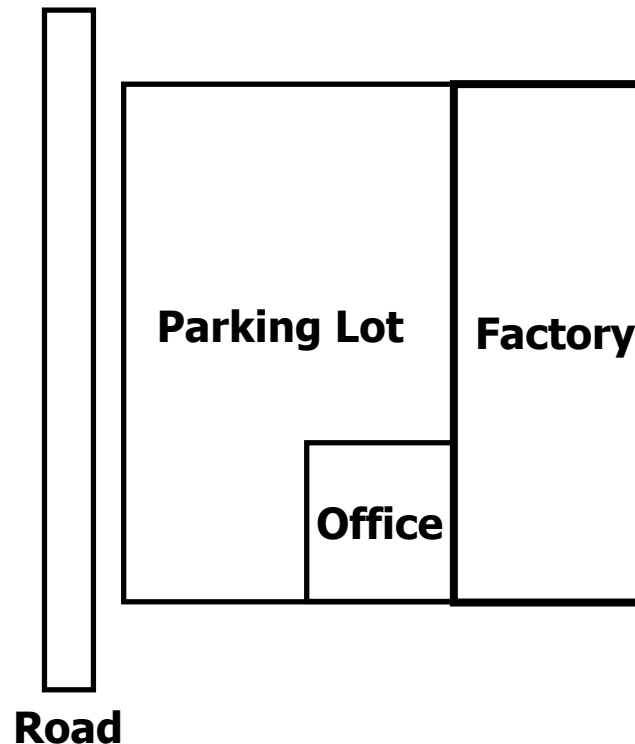
Orient Building for Acoustical Advantage



Orientation

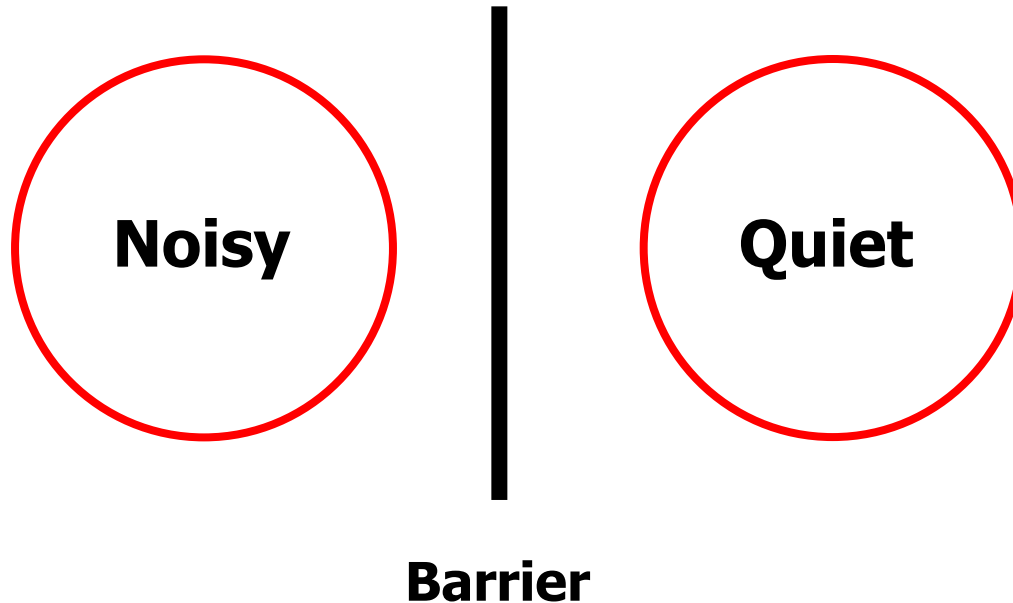
Orient Building for Acoustical Advantage

Note: Sound is 3-dimensional,
check overhead for
flight paths



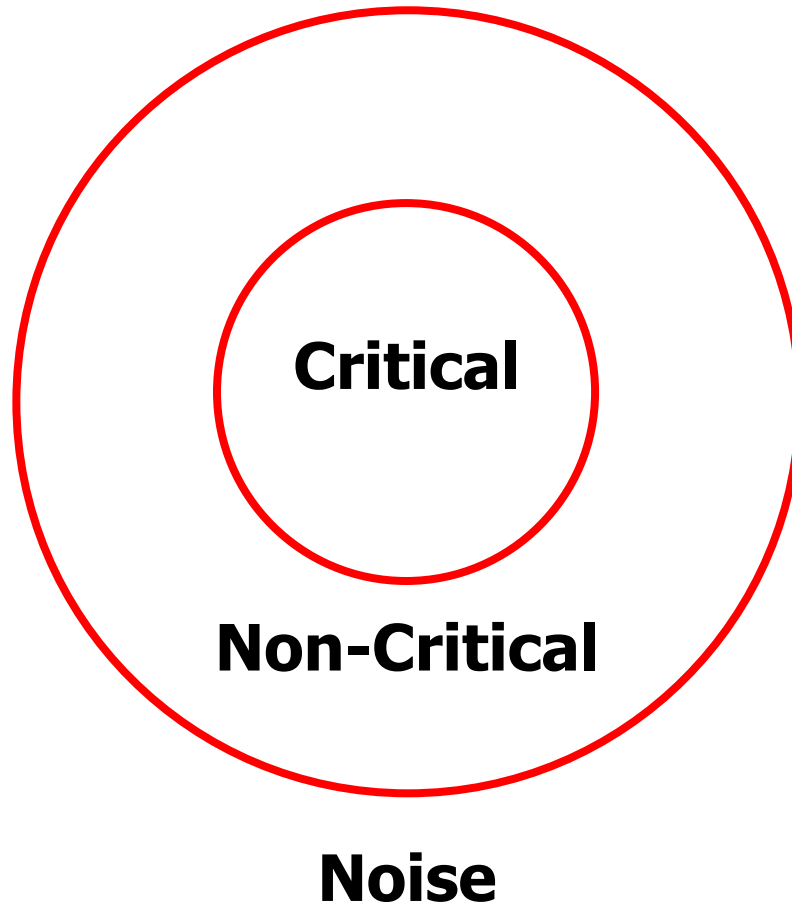
Planning

Consider Acoustical Sensitivity of Activities



Planning

Consider Acoustical Sensitivity of Activities



Internal Layout

Each room has needs that can be met by room layout

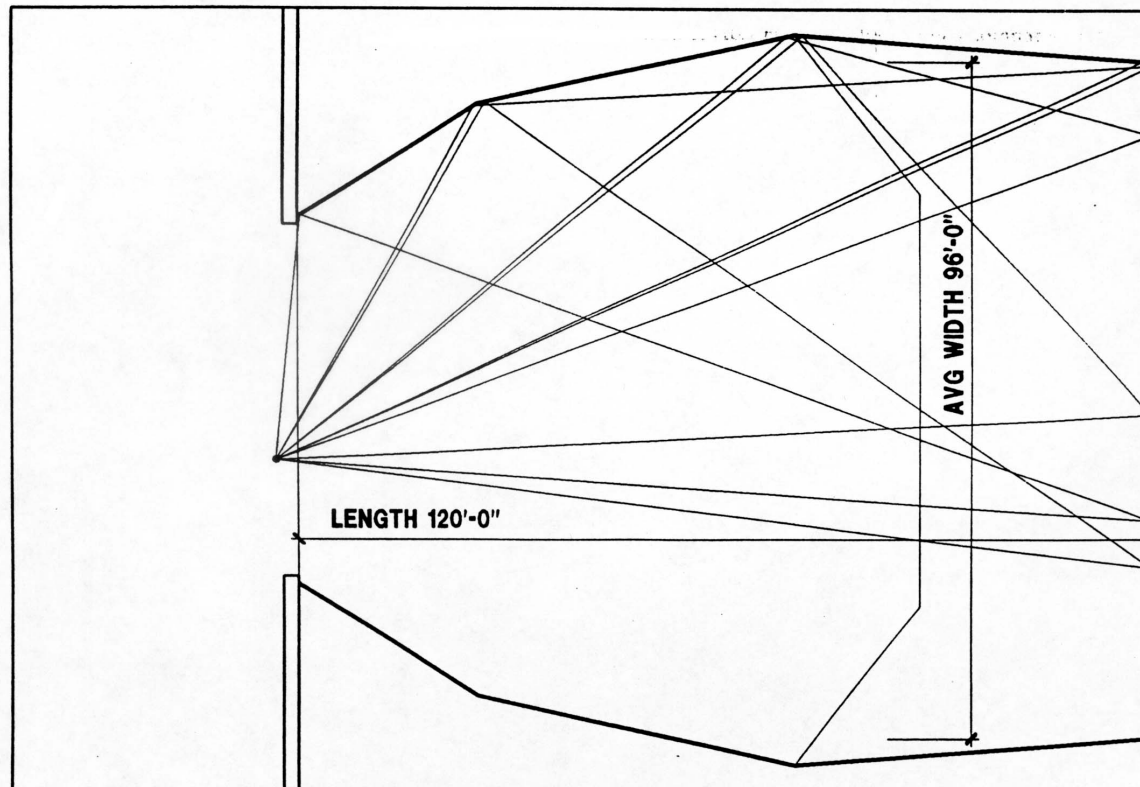


Figure 5-12. Auditorium exercise plan.

Basic Acoustic Goals

1. Provide adequate isolation
2. Provide appropriate acoustic environment
3. Provide appropriate internal function
4. Integrate 1-3 amongst themselves and into comprehensive architectural design

Sound

- Mechanical vibration, physical wave or series of pressure vibrations in an elastic medium
- Described in **Hertz** (cycles per second)
- Range of hearing: 20 - 20,000 hz

Noise

- **Any unwanted sound**

Sound Propagation

Sound travels at different speeds through various media.

<u>Media</u>	<u>Speed (C)</u>
Air:	1,130 fps
Water:	4,625 fps
Wood:	10,825 fps
Steel:	16,000 fps

Wavelength

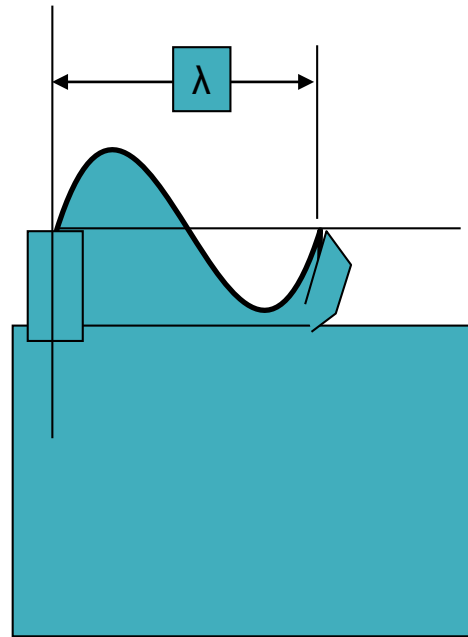
Distance between similar points on a successive wave

$$C = f \lambda \quad \text{or} \quad \lambda = C / f$$

C = velocity (fps)

f = frequency (hz)

λ = wavelength (ft)



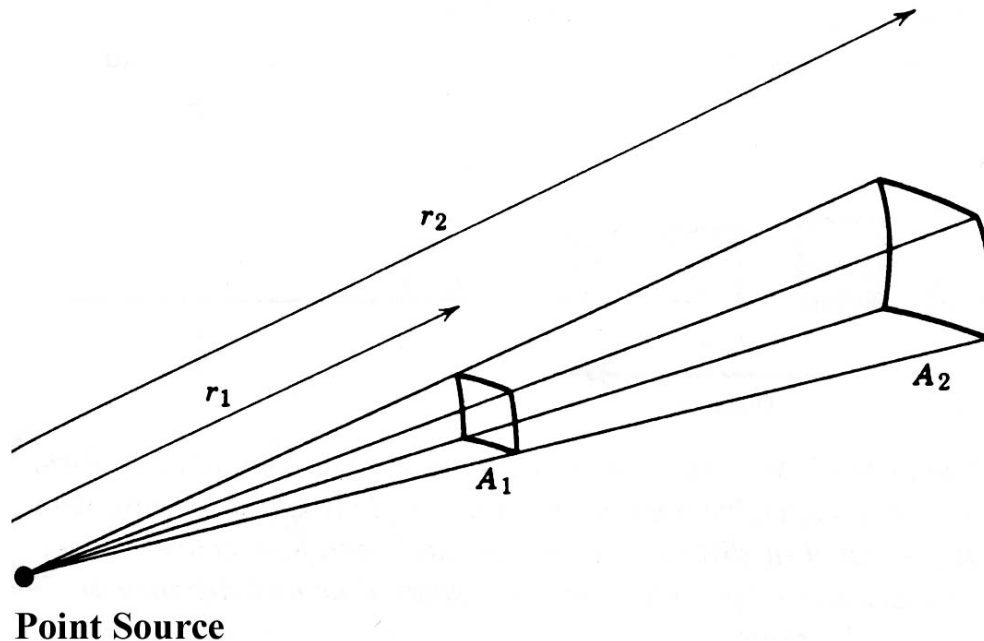
Lower frequency: longer wavelength

Sound Magnitude

- Sound Power (P)
- Sound Intensity (I)

Sound Power

- Energy radiating from a point source in space.
- Expressed as **watts**



Sound Intensity

Sound power distributed over an area

$$I = P / A$$

I: sound (power) intensity, W/cm²

P: acoustic power, watts

A: area (cm²)

Intensity Level

Level of sound relative to a base reference

Comparison of Decimal, Exponential, and Logarithmic Statements of Various Acoustic Intensities

<i>Intensity (W/cm²)</i>		<i>Intensity Level— Logarithmic Notation</i>	<i>Examples</i>
<i>Decimal Notation</i>	<i>Exponential Notation</i>		
0.001	10^{-3}	130 db	Painful
0.0001	10^{-4}	120 db	75-piece orchestra
0.00001	10^{-5}	110 db	
0.000001	10^{-6}	100 db	
0.0000001	10^{-7}	90 db	Shouting at 5 ft
0.000000001	10^{-9}	70 db	Speech at 3 ft
0.000000000001	10^{-11}	50 db	Average office
0.00000000000001	10^{-13}	30 db	Quiet unoccupied office
0.000000000000001	10^{-14}	20 db	Rural ambient
0.0000000000000001	10^{-15}	10 db	Threshold of hearing
0.00000000000000001	10^{-16}	0 db	

Intensity Level

Extreme range dictates the use of logarithms

$$IL = 10 \log (I/I_0)$$

IL: intensity level (dB)

I: intensity (W/cm²)

I₀: base intensity (10⁻¹⁶ W/cm², hearing threshold)

Log: logarithm base 10

Intensity Level Scale Change

Changes are measured in decibels

scale change

3 dB

6 dB

7 dB

subjective loudness

barely perceptible

perceptible

clearly perceptible

Note: round off to nearest whole number

Intensity Level—The Math

If $IL_1=60$ dB and $IL_2=50$ dB,
what is the total sound intensity?

1. Convert to intensity

$$IL_1 = 10 \log (I_1/I_0)$$

$$60 = 10 \log(I_1/10^{-16})$$

$$6.0 = \log(I_1/10^{-16})$$

$$10^6 = I_1/10^{-16}$$

$$I_1 = 10^{-10}$$

$$IL_2 = 10 \log (I_2/I_0)$$

$$50 = 10 \log(I_2/10^{-16})$$

$$5.0 = \log(I_2/10^{-16})$$

$$10^5 = I_2/10^{-16}$$

$$I_2 = 10^{-11}$$

Intensity Level—The Math

If $IL_1=60$ dB and $IL_2=50$ dB,
what is the total sound intensity?

2. Add together

$$I_1 + I_2 = 1 \times 10^{-10} + 1 \times 10^{-11}$$

$$I_{TOT} = 11 \times 10^{-11} \text{ W/cm}^2$$

Intensity Level—The Math

If $IL_1=60$ dB and $IL_2=50$ dB,
what is the total sound intensity?

3. Convert back to intensity

$$IL_{TOT} = 10 \text{ Log } (I_{TOT}/I_0)$$

$$IL_{TOT} = 10 \text{ Log } (11 \times 10^{-11}) / 10^{-16}$$

$$IL_{TOT} = 10 (\text{Log } 11 + \text{Log } 10^5)$$

$$IL_{TOT} = 10 (1.04 + 5) = 60.4 \text{ dB}$$

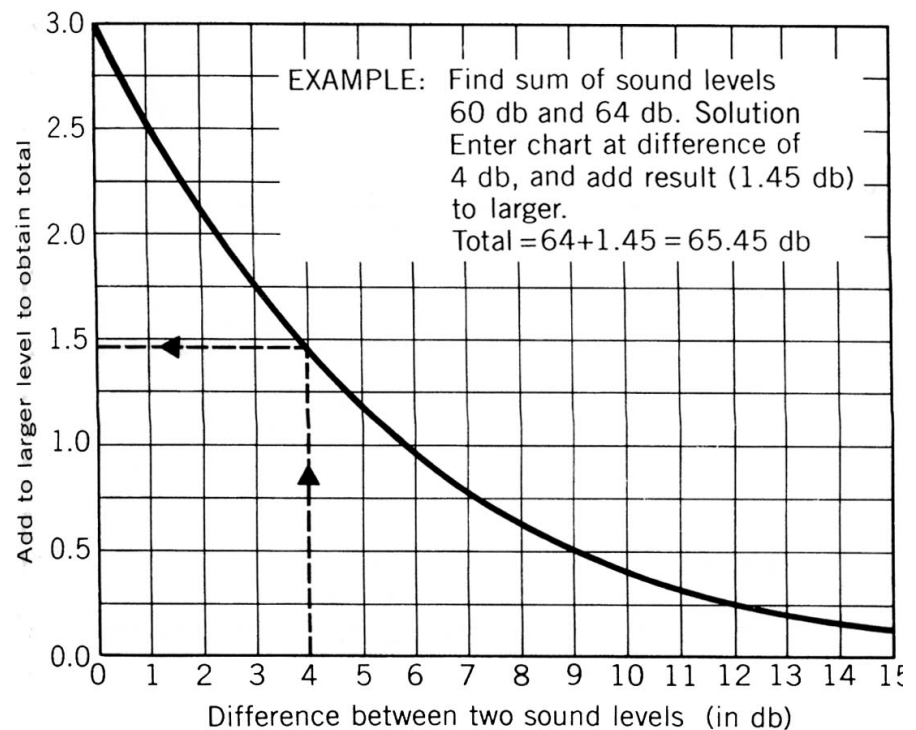
Intensity Level

- Add two 60 dB sources

$\Delta \text{dB} = 0$,

add 3 dB to higher

$\text{IL} = 60 + 3 = 63 \text{ dB}$



Sound Pressure Level

Amount of sound in an enclosed space

$$\text{SPL} = 10 \log (p^2/p_0^2)$$

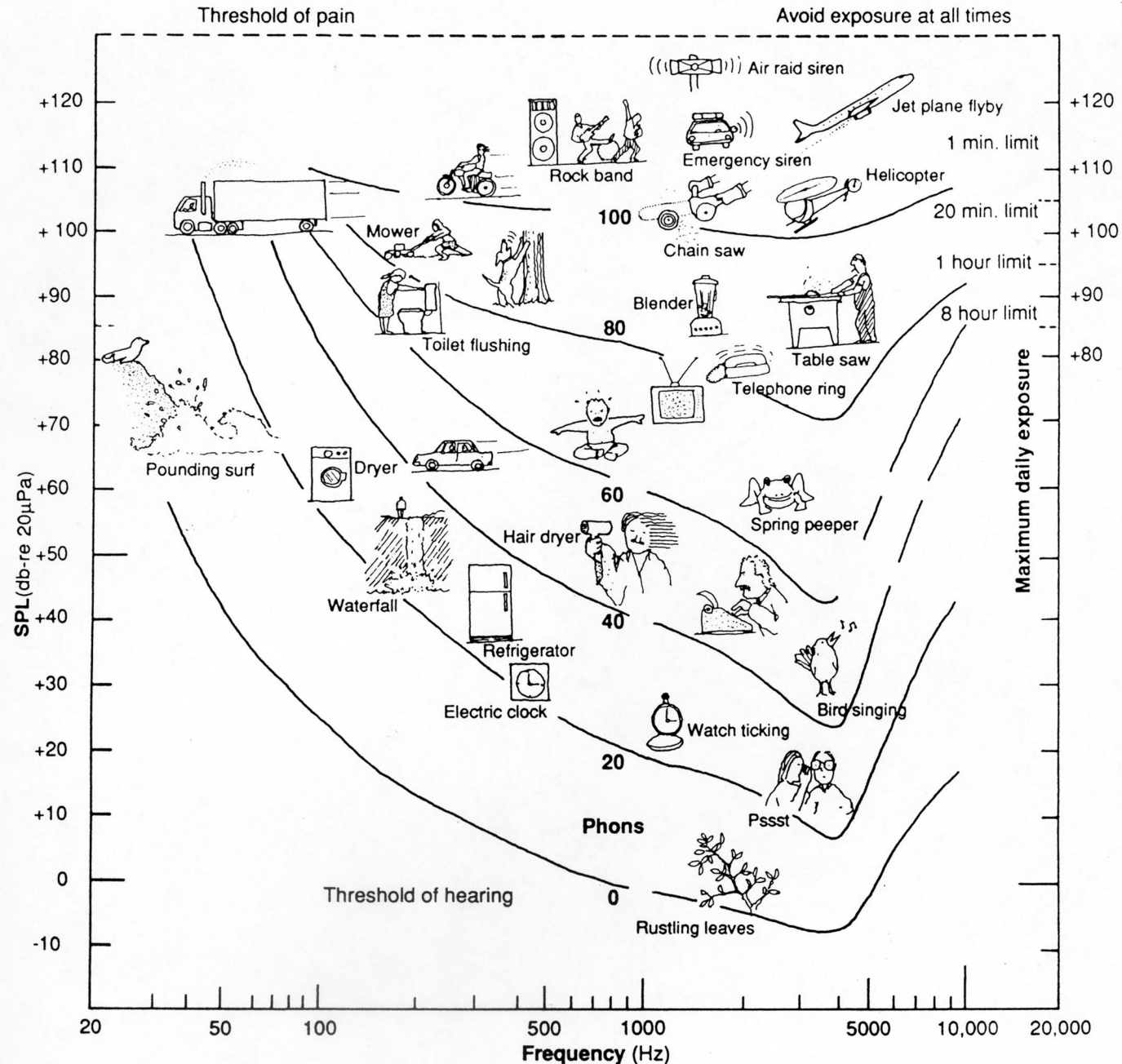
SPL: sound pressure level (dB)

p: pressure (Pa or μbar)

p_0 : reference base pressure (20 μPa or
2E-4 μbar)

Perceived Sound

Dominant frequencies affect sound perception



Sound Meter—"A" Weighting

Sound meters that interpret human hearing use an "A" weighted scale

dB becomes dBA

ARC4443 Acoustical, Electrical, and Illumination systems

Week 11-2

Architectural Acoustics

Design: Case Studies

COMPUTATIONS AND TECHNICAL REQUIREMENTS

CALCULATING THE ACOUSTICAL ENVIRONMENT
> 215 people X 3.1 cu.m = 666.5 cu.m
> Using 70% --> 666.5 cu.m /.7 = **952 cu.m approx**

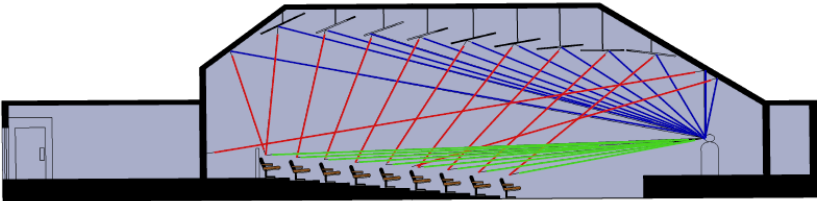
FIND OPTIMUM REVERB. TIME
> OptRT= 0.30log(952/10) = **0.593 secs approx**

FIND TARGET GOAL FOR ABSORPTION
> Using: TargetRT = (0.16 x V)/A --> Find "A"
> A = (0.16 x 952)/.593 = **256.9 sabins**

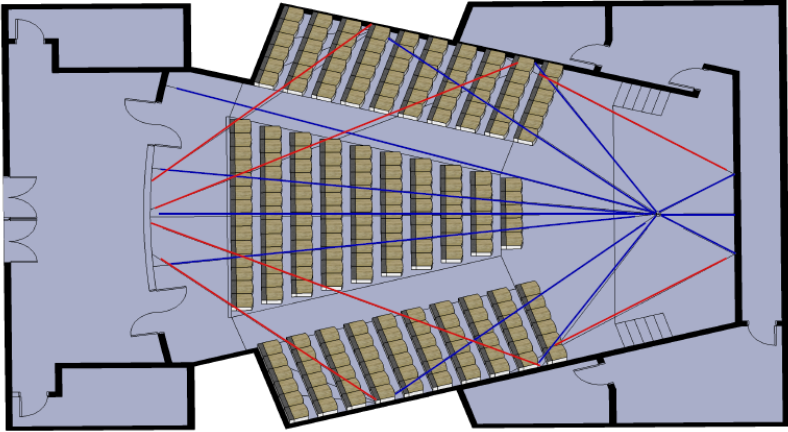
MATERIALS SPECS & ABSORPTION TABLE

MATERIAL	UNIT NRC	QUANTITY(sq.m)	ABSORPTION	LOCATION
Seats (Upholstered, Occ)	0.45	172	77.4	Seating Area
Seats (Upholstered, Unocc.)	0.3	43	12.9	Seating Area
Carpet on Concrete	0.15	181.513	27.22695	Audience Flr.
Painted Concrete	0.08	66	5.28	Stage Wall
Perforated Panel 4" Airspace	0.8	56.811	45.4488	Rear Wall
T&G Hardwood	0.1	98.7	9.87	Stage Floor
Suspended Acoustical Board	0.3	148	44.4	Mid Ceiling
Painted Concrete	0.08	37.138	2.97104	Stage Ceiling
Suspended Acoustical Board	0.45	24.878	11.1951	Rear Ceiling
4mm Plywood Diffusers	0.3	50.476	15.1428	Side Walls
T&G Hardwood	0.1	50.242	5.0242	Side Walls
			256.85889	
			TOTAL ABSORPTION	

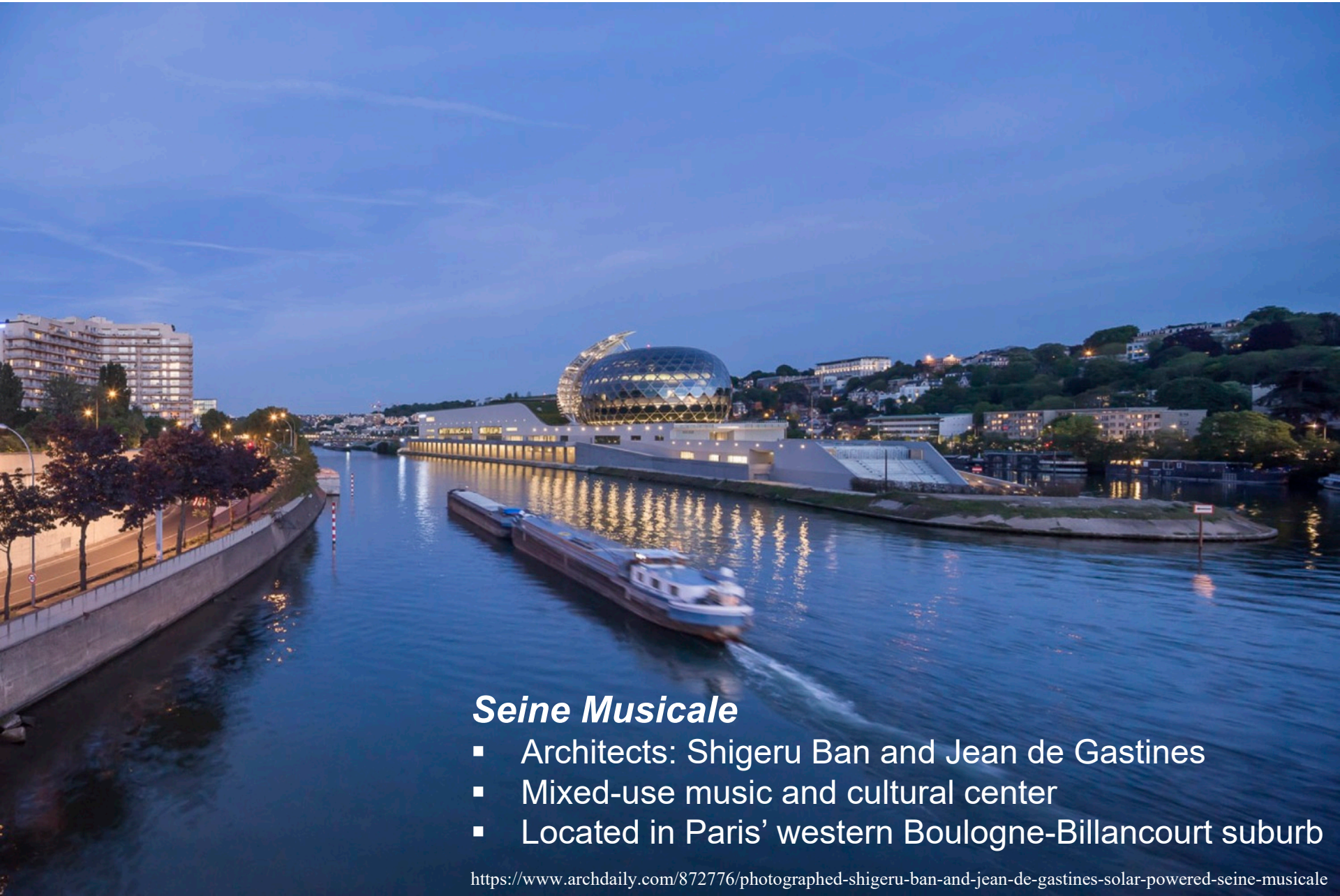
GEOMETRIC ACOUSTICS DIAGRAMS (1:100m)



IN SECTION.



IN PLAN.



Seine Musicale

- Architects: Shigeru Ban and Jean de Gastines
- Mixed-use music and cultural center
- Located in Paris' western Boulogne-Billancourt suburb

<https://www.archdaily.com/872776/photographed-shigeru-ban-and-jean-de-gastines-solar-powered-seine-musicale>







- Foyer (Public reception area)
- Classical music concert hall "Auditorium"
- Multi purpose concert hall "Grande Seine"
- Choir school "Maîtrise des Hauts-de-Seine"
- Resident orchestra area
- Rehearsal and recording area
- Business and event area
- Restaurants and retail
- Day nursery

ShigeruBanArchitects Europe + Jean de Gastines Architectes // La Seine Musicale

Section A-A' // S=1/1000

2017-08-27

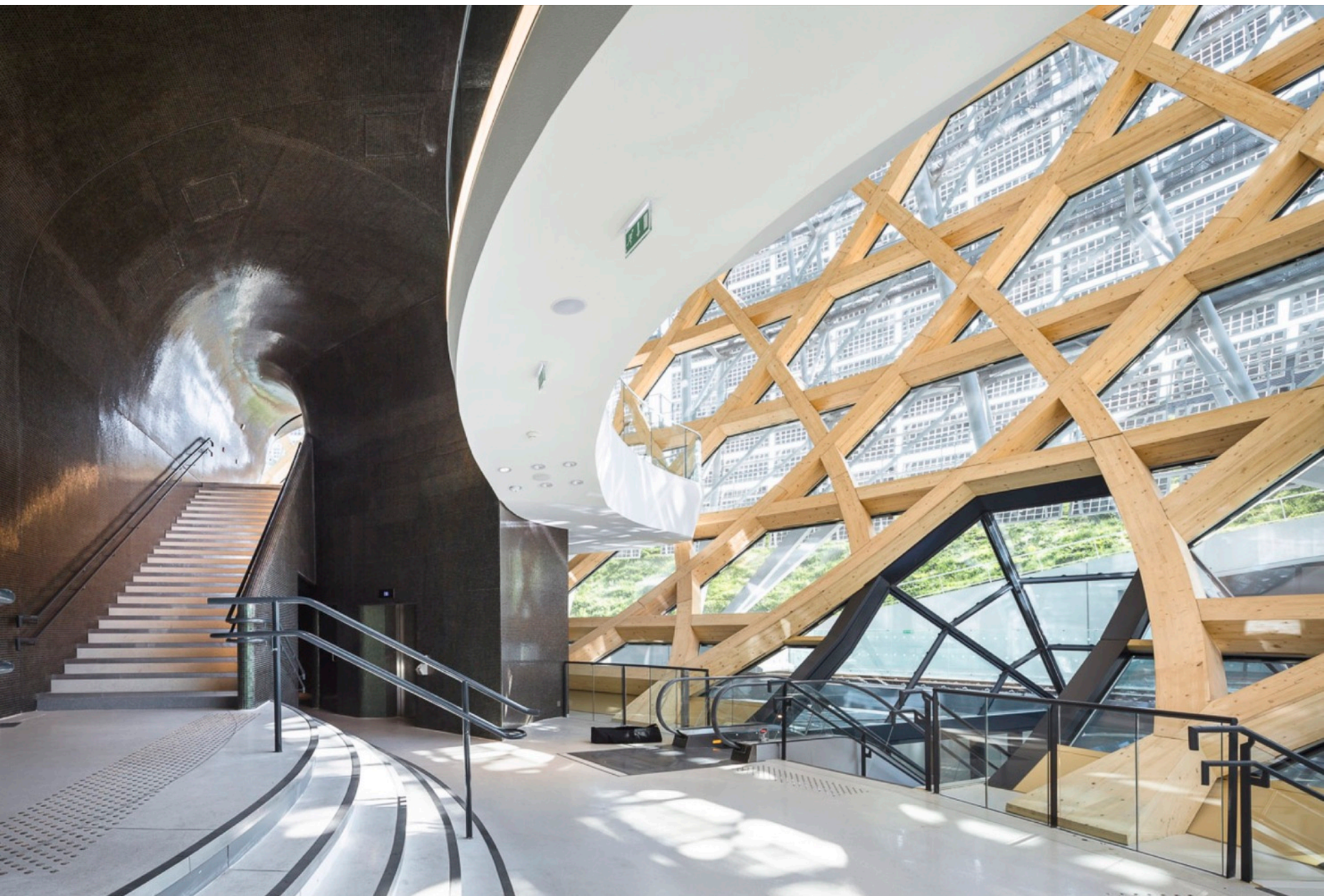
Section. La Seine Musicale by Shigeru Ban Architects Europe + Jean de Gastines Architectes.

Image 36 of 39

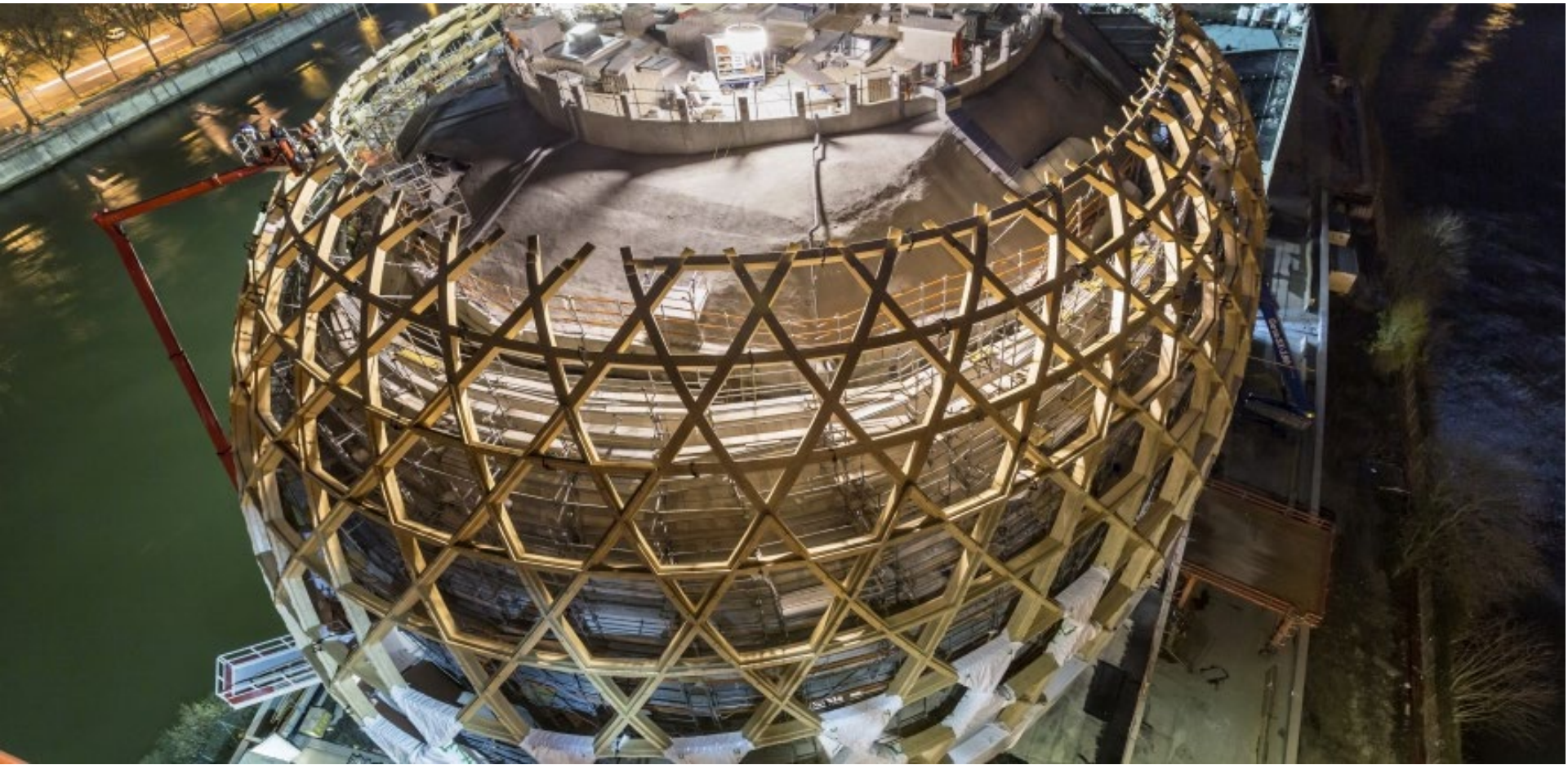


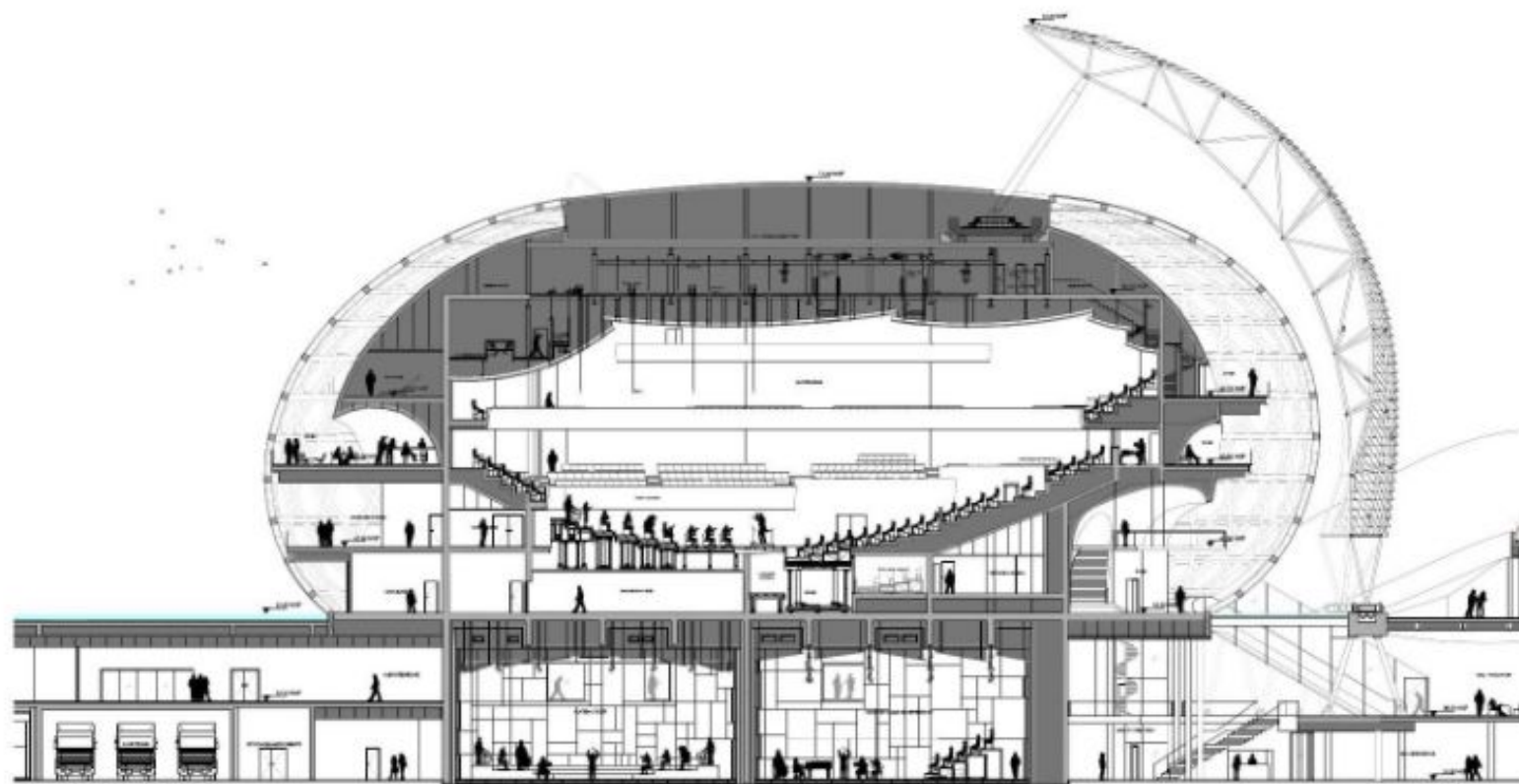




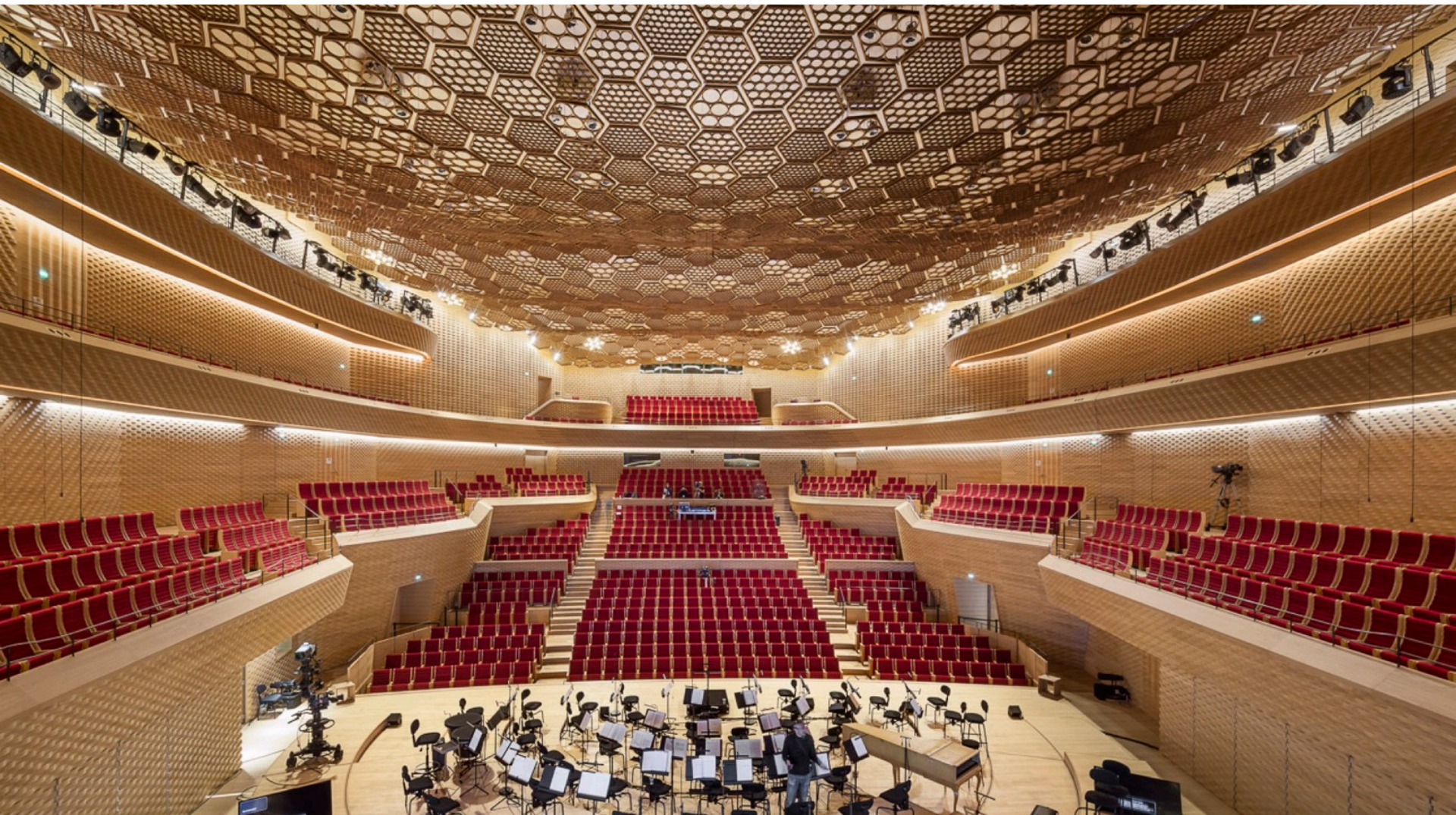








Auditorium, section. La Seine Musicale by Shigeru Ban Architects Europe + Jean de Gastines Architectes.
Image 33 of 39

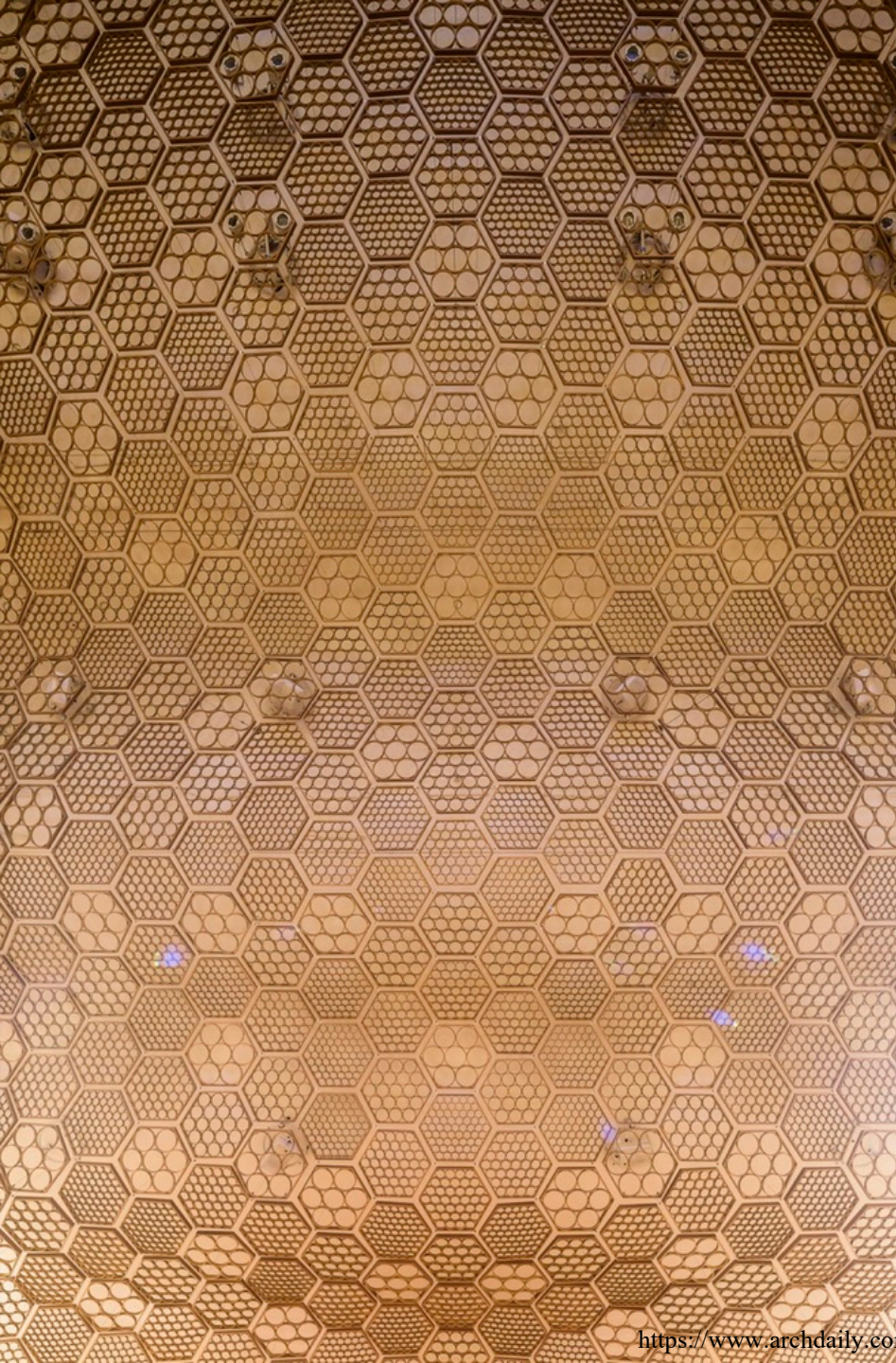


Inside the music hall, you will find uniquely fabricated walls and ceilings: walls are furnished with corrugated plywood panels that are laid and woven irregularly to optimize the acoustic effect. Ceilings are undulated using Ban's signature material: paper tubes....

<http://www.interactiongreen.com/shigeru-ban-projects-progress/>



<https://www.archdaily.com/7872776/photographed-shigeru-ban-and-jean-de-gastines-solar-powered-seine-musicale>



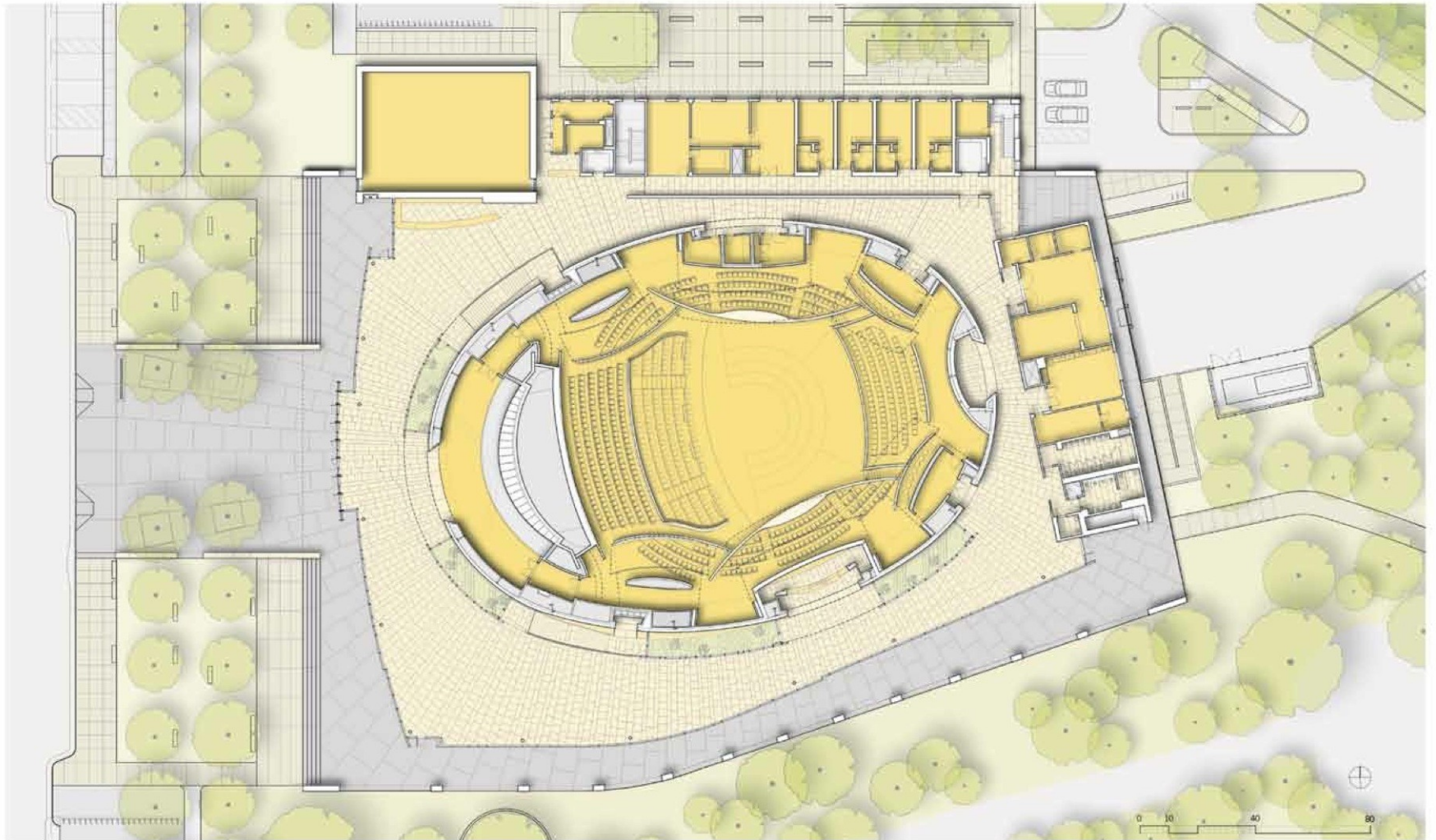


Architects: Ennead Architects

Location: Lasuen Street, Stanford University,
Stanford, CA, US

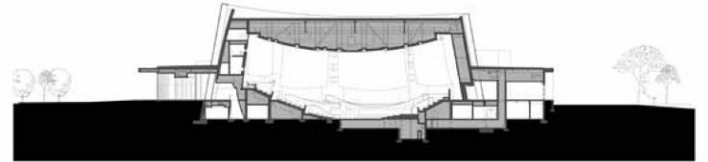
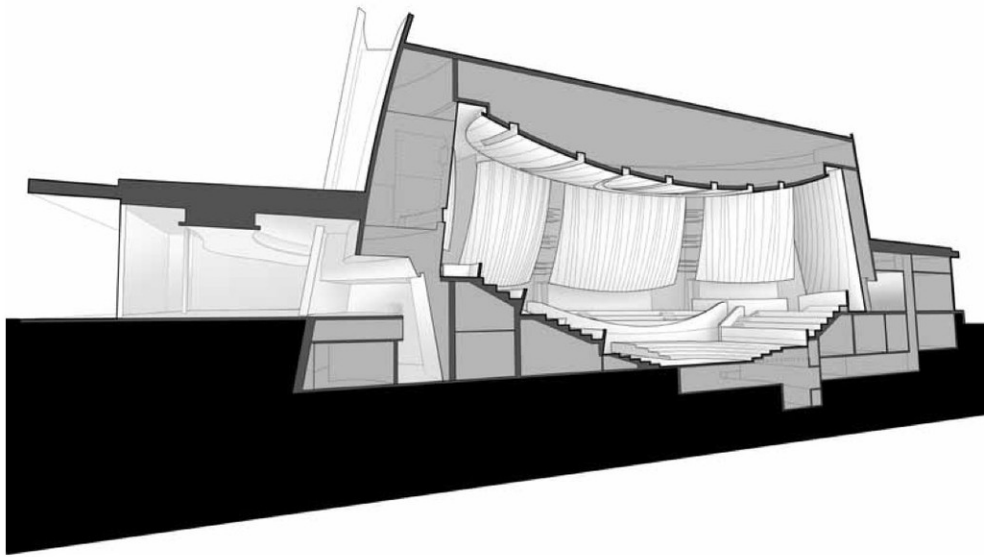
Area: 112,365.0 sqm

Project Year: 2012

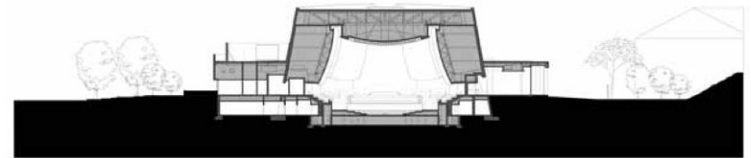
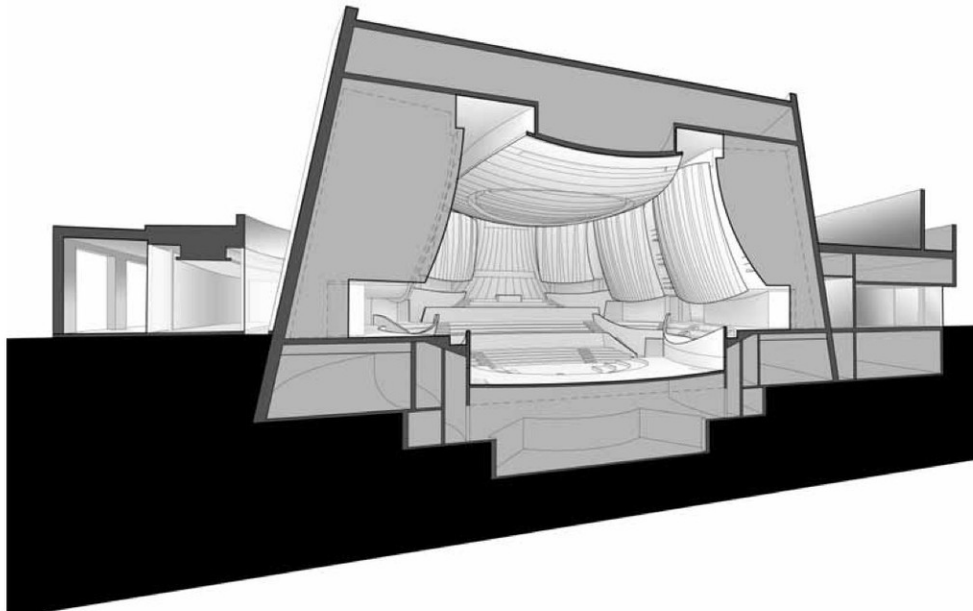


LOBBY LEVEL PLAN

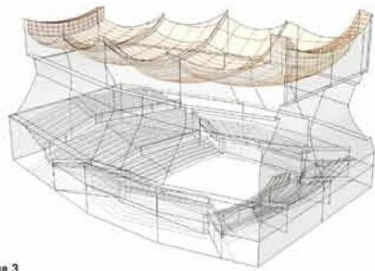




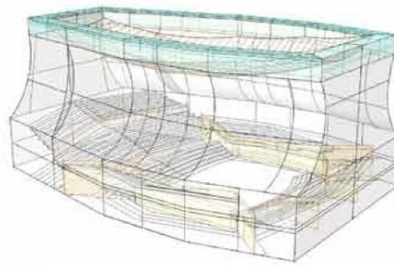
LONGITUDAL SECTION



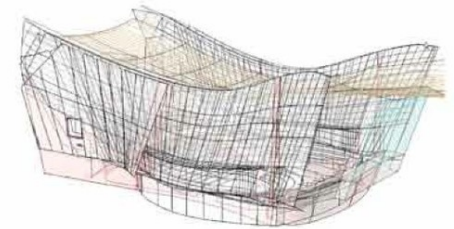
TRANSVERSE SECTION



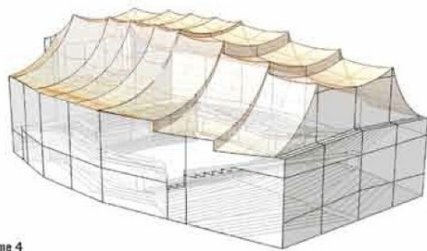
scheme 3



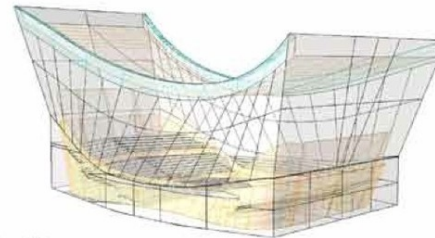
scheme 11



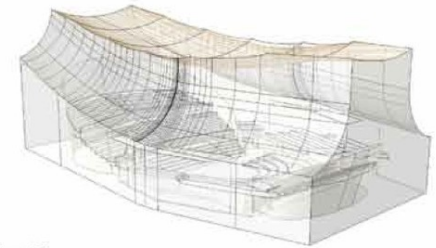
scheme 15



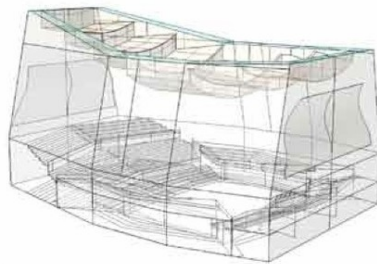
scheme 4



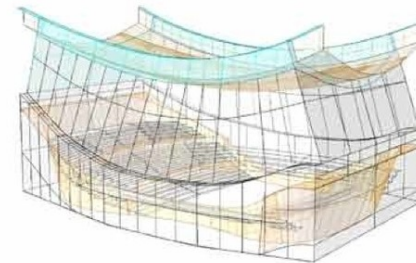
scheme 12b



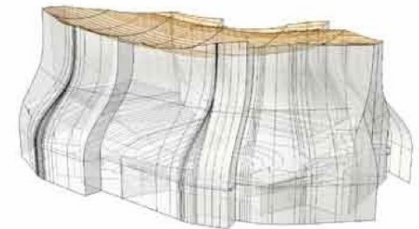
scheme 16



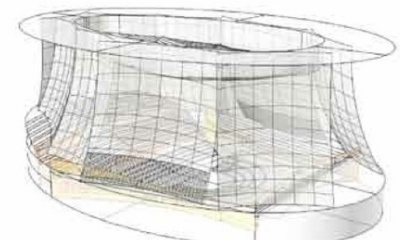
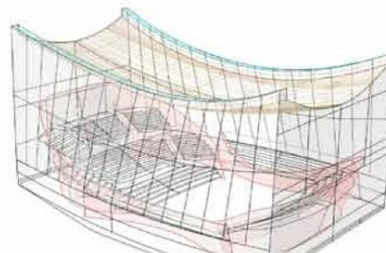
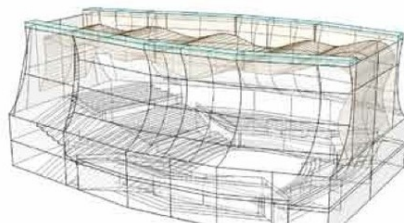
scheme 6b



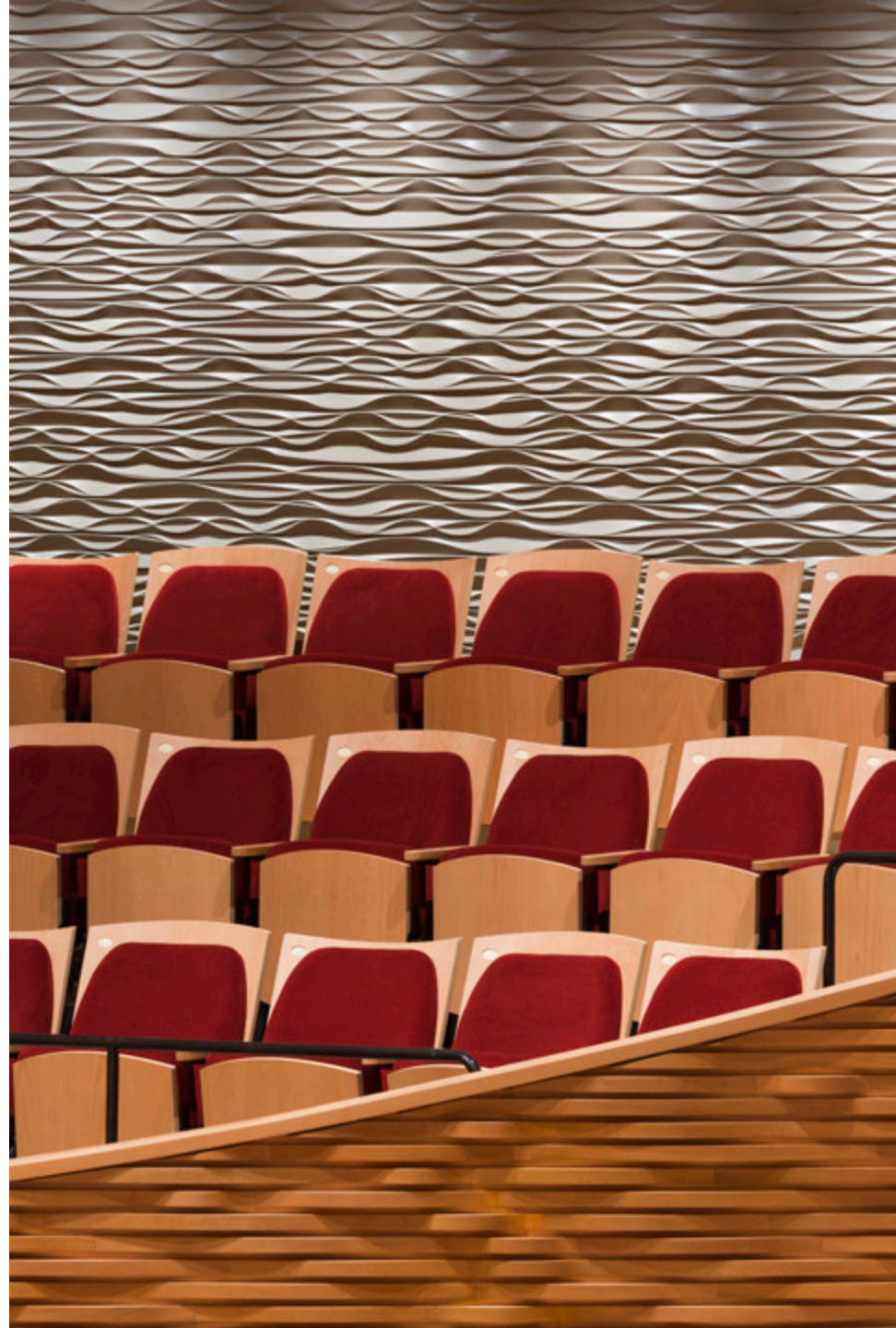
scheme 12c



scheme 17



CONCEPT ACOUSTIC MODEL STUDIES



Reference

- M. David Egan, Architectural Acoustics (Acton, MA: Copley Custom Textbooks, 2008) ISBN 1-58152-507-9
- Acoustics 1 Lecture slides, College of Architecture + Planning, University of Utah
- www.archdaily.com
- Mechanical & Electrical Equipment for Buildings (MEEB), 12th Edition, Walter T. Grondzik and Alison G. Kwok, John Wiley and Sons, Inc., 2015, ISBN: 978-1-118-61690-4